Part 1 : CHOOSE THE CORRECT ANSWER (24 POINTS)

الاجابة تكون في الجدول الموجود باسفل صفحة الغلاف

1. If computed values are close to each others but far away from the true value, then we have;
   A. high accuracy and high precision
   B. low accuracy and high precision
   C. low accuracy and low precision
   D. high accuracy and low precision

2. The solution of the equation $ax^2 + bx + c = 0$ could be written as $x =$
   A. $\dfrac{-2c}{b \pm \sqrt{b^2 - 4ac}}$
   B. $\dfrac{-2a}{b \pm \sqrt{b^2 - 4ac}}$
   C. $\dfrac{2c}{b \pm \sqrt{b^2 - 4ac}}$
   D. $\dfrac{2a}{b \pm \sqrt{b^2 - 4ac}}$

3. Use modified Newton-Raphson method to approximate the root of the equation;
   $x^3 - 2x^2 - 4x + 8 = 0$, if the initial guess $x_0 = 1$, then $x_3 =$
   A. 1.789473684
   B. 1.95959596
   C. 1.998836672
   D. 1.999791764

4. Approximate 12.478562149 to 4 significant digits by chopping ?
   A. 12.47 B. 12.48 C. 12.4785 D. 12.4786

5. Normalize; 0.00000027×10^m, where m is an integer ?
   A. $0.27 \times 10^{m+3}$ B. $0.27 \times 10^{m-3}$ C. $0.27 \times 10^{m+6}$ D. $0.27 \times 10^{m-6}$

6. Given that the equation $f(x) = 0$, has a root in the interval $[2, 2.5]$. Applying the bisection method to find this root, if the tolerance equals 0.00001, then the least number of iterations you should do is:
   A. 14 B. 13 C. 16 D. 15

7. Given that 1 is a root of the equation;
   $x^3 - 3x^2 + 3x - x^0 = 0$
   then the multiplity of this root equals:
   A. 1 B. 3 C. 2 D. 4

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8. Use bisection method twice to approximate the root of

$$\frac{20}{3+x} = 3+\sqrt{x} \quad , x \in [1,2]$$

A. 1.25 B. 0.75 C. 1.75 D. 0.25

9. Rewriting the equation;

$x^3 + 4x - 4 = 0$ in the form $x = a + bx^3$ then the orderd pair $(a,b) =$

A. $(\frac{1}{4},1)$ B. $(-\frac{1}{4},1)$ C. $(1,\frac{1}{4})$ D. $(1,-\frac{1}{4})$

10. Given that $x_{i+1} = \frac{1}{5}\sec^2 x_i$, $x_0 = 0.5$.

Find x_4 ? (Round your answer to 4 decimal places)

A. 0.1716 B. 0.2000 C. 0.2091 D. 0.1717

11. Use three non-zero terms of Maclaurin series corresponding to the function $f(x) = \frac{1}{1+x^4}$

to approximate the value of the integral $\int_0^{0.4} \frac{dx}{1+x^4}$? (Round your answer to 6 decimal places)

A. 0.585476 B. 0.746145 C. 0.749377 D. 0.585568

12. If the estimated height of a tower is 46 m , if it's true length is 50 m then the pecent relative true error equals:

A. 8% B. 12% C. 10% D. 14%

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3) The volume of a cone (مخروط) is $10cm^3$, if its height; $h = 6\ cm$. Use Newton-Raphson method twice to approximate its radius, take the initial guess to be 1. " Round off your answer to six decimal places"

Hint: the volume of the cone formula $(V = \frac{1}{3}\pi r^2 h)$ (4 Points)

$V = 10$

$10 = \frac{1}{3}\pi r^2 h$

$10 = \frac{1}{3}\pi r^2 (6) \Rightarrow 0 = 2\pi r^2 - 10$

$\frac{30}{6\pi} = r^2$

$\sqrt{\frac{5}{\pi}} = r$

$1.261566 = r$

true radius

$f(r) = 2\pi r^2 - 10$

$f'(r) = 4\pi r$

$r_1 = r_0 - \frac{2\pi r^2 - 10}{4\pi r}$ $\quad r_0 = 1$

$r_1 = 1.2957747715$

$r_2 = 1.262017813$

$r_2 \approx 1.262018$

6 d.p

4

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